10, A police jeep, approaching an orthogonal intersection from the northern direction, is chasing a speeding car that has turned and moving straight east. When the jeep is 0.6 km north of the intersection and the car is 0.8 km to the east. The police determine with a radar that the distance between them and the car is increasing at 20 km/hr. If the jeep is moving at 60 km/hr at the instant of measurement, what is the speed of the car?

7.2.4 Equations of Tangent and Normal

According to Leibniz, tangent is the line through a pair of very close points on the curve.

Definition 7.1

The tangent line (or simply tangent) to a plane curve at a given point is the straight line that **just touches** the curve at that point.

Definition 7.2

The normal at a point on the curve is the straight line which is perpendicular to the tangent at that point.

The tangent and the normal of a curve at a point are illustrated in Fig. 7.7.

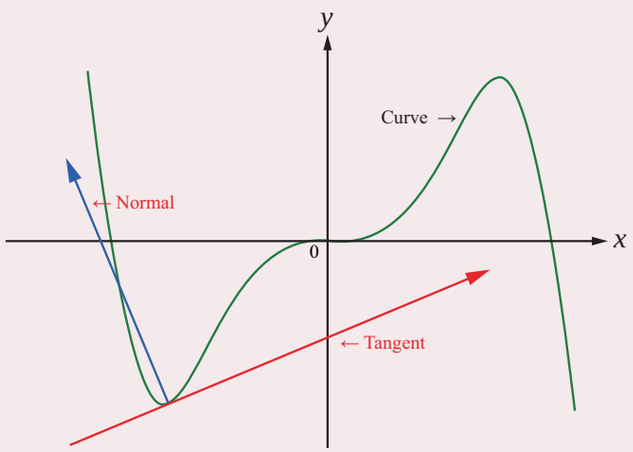


Fig.7.7

Consider the given curve $y = f(x)$.

The equation of the tangent to the curve at the point, say (a,b), is given by

$$y - b = (x - a) \times \left(\frac{dy}{dx}\right)_{(a,b)} \quad \text{or} \quad y - b = f'(a) \cdot (x - a).$$

In order to get the equation of the normal to the same curve at the same point, we observe that normal is perpendicular to the tangent at the point. Therefore, the slope of the normal at (a,b) is the negative of the reciprocal of the slope of the tangent which is $-\left(\frac{1}{\frac{dy}{dx}}\right)_{(a,b)}$.

Hence, the equation of the normal is ,

$$(y - b) = -\left(\frac{1}{\frac{dy}{dx}}\right)_{(a,b)} \times (x - a) \quad \text{or} \quad (y - b) \times \left(\frac{dy}{dx}\right)_{(a,b)} = -(x - a).$$

Remark

(i) If the tangent to a curve is horizontal at a point, then the derivative at that point is 0. Hence, at that point (x_1, y_1) the equation of the tangent is $y = y_1$ and equation of the normal is $x = x_1$.

(ii) If the tangent to a curve is vertical at a point, then the derivative exists and infinite (∞) at the point. Hence, at that point (x_1, y_1) the equation of the tangent is $x = x_1$ and the equation of the normal is $y = y_1$.

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